



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech IV Semester

End Semester Examination - April / May -2025

Design and Analysis of Algorithm [ECM-2224]

Marks: 50

Duration: 180 mins.

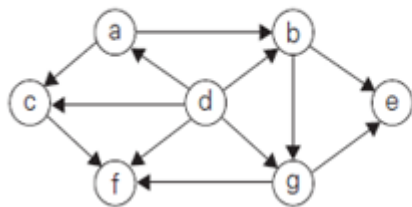
PART-A

Answer all the questions.

- 1) List the general plan for analyzing time efficiency of recursive algorithms. Set up and solve the recurrence relation for the number of moves made by Tower of Hanoi algorithm using backward substitution method (4)
 - A)
 - B) For each of the following recurrences, give an expression for the runtime $T(n)$ if the recurrence can be solved with the Master Theorem. Otherwise, indicate that the Master Theorem does not apply. (3)
 1. $T(n) = 3T(n/2) + n^2$
 2. $T(n) = T(n/2) + 2^n$
 3. $T(n) = 2^n T(n/2) + n^n$
 - C) Write a Pseudocode of Binary search Algorithm and calculate its time complexity with all necessary Steps. (3)
- 2) What are the advantages of Circular Queue over Queue ? Write a menu drive program to implement simple arithmetic operation in Circular Queue (4)
 - A)
 - B) Convert the following infix expression into its equivalent postfix expression. (3)
$$A + (B * C (D / E F) * G) * H$$
 - C) Define stacks. List and explain the various operations on stacks using arrays with stack overflow and stack underflow conditions. (3)
- 3) Given the array 70, 60, 45, 30, 20, 10, 5, 15, 25, 35, construct a **max-heap** using the **bottom-up heap construction technique**.
 - A)
 1. Show the steps of heapification.
 2. What is the total number of comparisons made during construction? (4)

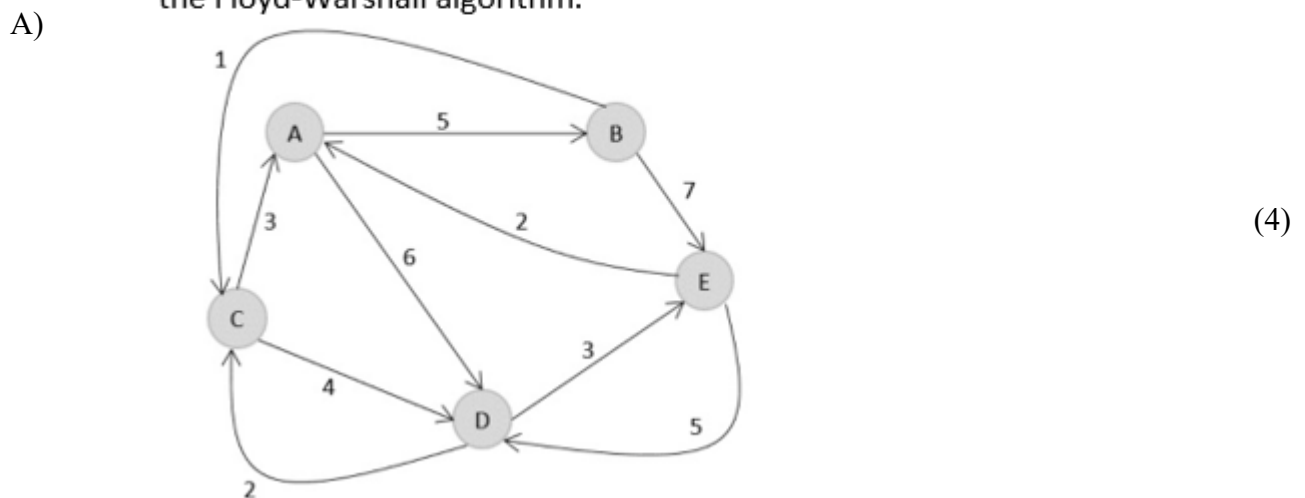
i) Analyze the worst-case time complexity and explain why the bottom-up approach is more efficient than repeated insertions.
 - B) (3)

Apply the source-removal algorithm to solve the topological sorting problem for the following digraph:



- C) Design a 2-3 tree for the list 18, 32, 12, 23, 30, 10, 15, 20, 21, 24, 31, 40, 45, 47, 50, 52 using successive insertion, showing all stages. (3)

- 4) Consider the following directed weighted graph $G = \{V, E\}$. Find the shortest paths between all the vertices of the graphs using the Floyd-Warshall algorithm.



- B) (a) Construct the Huffman tree and obtain the Huffman code for the data given in Table.

Symbol	A	E	H	S	W
Frequency	0.2	0.2	0.15	0.3	0.15

(4)

Decode 11100011100101110100 using the code obtained in (a)

- C) Apply the Bottom up Dynamic Programming Technique to the following instance of Knapsack problem with Capacity $W = 5$.

item	weight	value
1	2	\$12
2	1	\$10
3	3	\$20
4	2	\$15

capacity $W = 5$.

(2)

- 5) (4)

A)

When can you call a graph as a minimum spanning tree? Apply Prim's algorithm to the graph shown in Figure 5A. Include in the priority queue all the vertices not already in the tree. List the edges included in the minimum spanning tree thus obtained. Write the total cost of the minimum spanning tree.

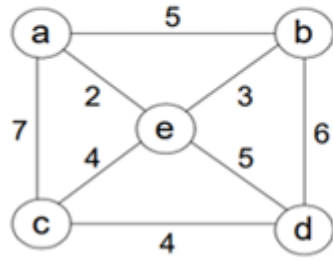


Figure 5A

B)

Demonstrate P, NP and NP complete complexity classes.

(3)

C)

Given 4 x 4 chessboard, arrange four queens in a way, such that no two queens attack each other.

(3)

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